

DATASCI 530 – Assignment 5

For this assignment you will practice the use of some of the machine learning methods we have used in class so far. We will be working with the “iris” dataset which contains information about different flower characteristics. You can download the data [here](https://archive.ics.uci.edu/dataset/53/iris).

<https://archive.ics.uci.edu/dataset/53/iris>

Your goal is to predict the label “class”.

- Start off with a description of the dataset (how many rows, features, what is the proportion of observations in each class, etc.)
- Run a comparison of different machine learning methods, that should include SVMs, random forests, neural networks, and at least one method we haven’t covered in class (lasso, naïve bayes, ridge, etc.)
 - o Compute the cross-validated error. Consider more than one evaluation metric.
 - o Run the default tuning parameters for all methods except one (you choose). For this method, define a grid of tuning parameters and find the optimum.

Make sure to properly document your approach. Give a recommendation about which method you think better predicts the data.

Deliverable

Submit a Jupyter notebook with you results on Canvas. Include markdown chunks to write comments on your findings and explain your methodology.

When submitting your work imagine that you are sharing this with colleagues at work, who have asked you to explain and justify key decisions and findings of your analysis. The file should be professionally presented, self-explanatory, and have clearly labeled sections.

Rubric:

You will be graded both on the code and the writing using the following point system:

Code	4 points
- Code runs without issues - Code is appropriate to answer question	

- File is appropriately organized, with different selections and comments clarifying the definition of variables and dataset names.	
Description of methodology - Explain key coding decisions via markdown chunks - Clarify any choices about how to clean data - Shows evidence of conducting quality checks (count sample size of merged datasets, check for the presence/absence of missing values, and/or compute any relevant descriptive statistics)	2 points
Results and Visualization - Write a short summary of the findings. - Correct interpretation of results and easy to follow - Includes graphs to display results from different groups - All graphs look professional (have axis and title labels, suitable color scheme, font sizes, legend, etc.)	4 points